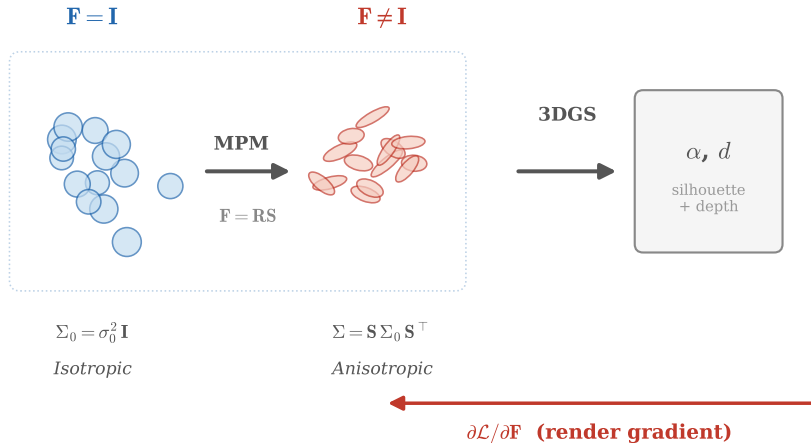


MPM particle $\equiv$ 3D Gaussian primitive (1:1 duality)



Gradient flow comparison

	Stretch $\nabla_s \mathcal{L}$	Rotation $\nabla_q \mathcal{L}$
$\mathbf{F} = \mathbf{I}$	$\neq 0$	$= 0$
$\mathbf{F} \neq \mathbf{I}$	$\neq 0$	$\neq 0$

F-derived covariance enables

rotation gradient channel